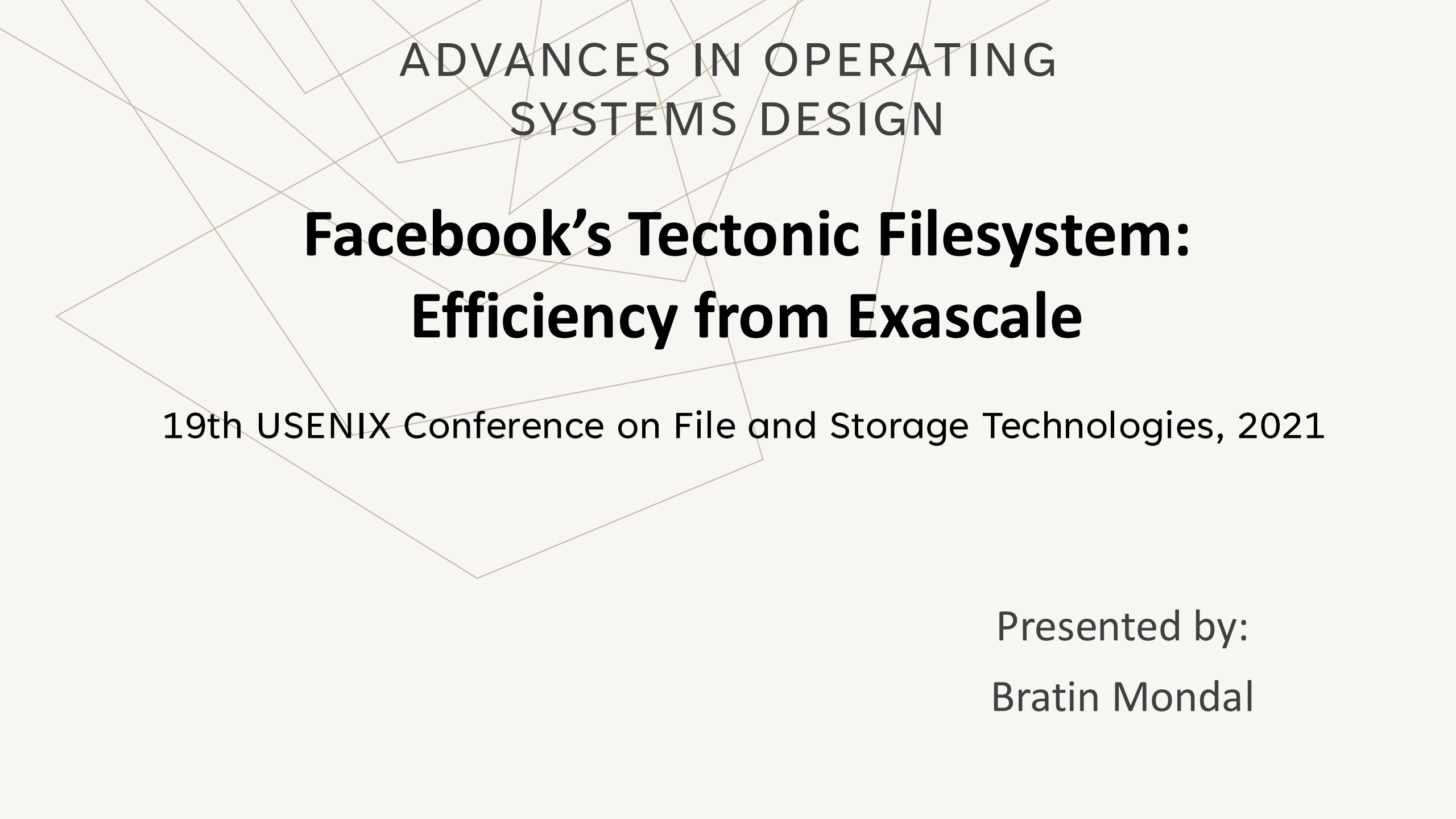




ADVANCES IN OPERATING  
SYSTEMS DESIGN

# **Facebook's Tectonic Filesystem: Efficiency from Exascale**

19th USENIX Conference on File and Storage Technologies, 2021

Presented by:  
Bratin Mondal

# Overview: What is Tectonic ?

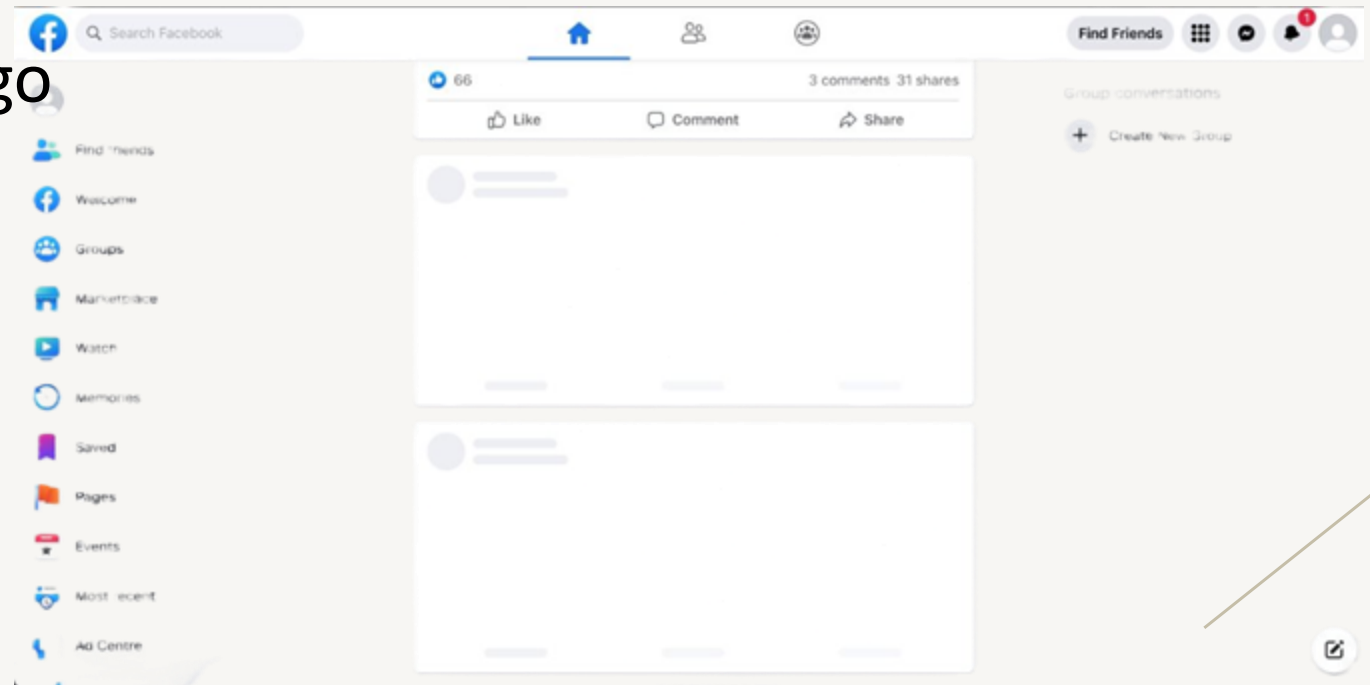
- Meta(Facebook)'s **exabyte-scale** distributed filesystem, append-only
- Serves around **ten tenants** with different requirements
  - Tenants are distributed systems that do not share data with each other and having **different set of requirements**
    - User media Storage (for serving user requests)
    - User history storage (for training analytics model)
- Not optimized for any particular workload but designed in such a way that **tenants can enable their optimizations**

# Meta's Storage Infrastructure before Tectonic

- Tenants used specialized storage systems
- Majorly two types of tenants
  - Blob Storage
  - Data Warehouse

# Blob (Binary Large Objects) Storage

- Immutable, varies from kilobytes(small photos) to megabytes(HD Videos)
- Low-latency reads and writes required
- Hot Blobs
  - Photo uploaded one minute ago
- Warm Blobs
  - Photo uploaded one year ago



# Haystack – Hot blob storage

- **Issue?**
  - Haystack needed very high IOPS
  - Need more devices to meet the IOPS requirements
  - Spare disk storage capacity
  - Replication factor **5.3x** in practice (Ideal – 3.6)
- **Can we use the surplus storage for others?**

- 500 GB storage, 8000 IOPS
- 8 Devices
- What do we do with the extra 300 GB?



Why don't we buy devices that meet our requirements more specifically?

# f4 – Warm blob storage

- **Issue?**
  - Large number of storage nodes needed to store the vast amount of warm blobs
  - But it also provides IOPS capacity which is not needed for the less frequently-accessed data
- **Can we use the surplus IOPS for others?**

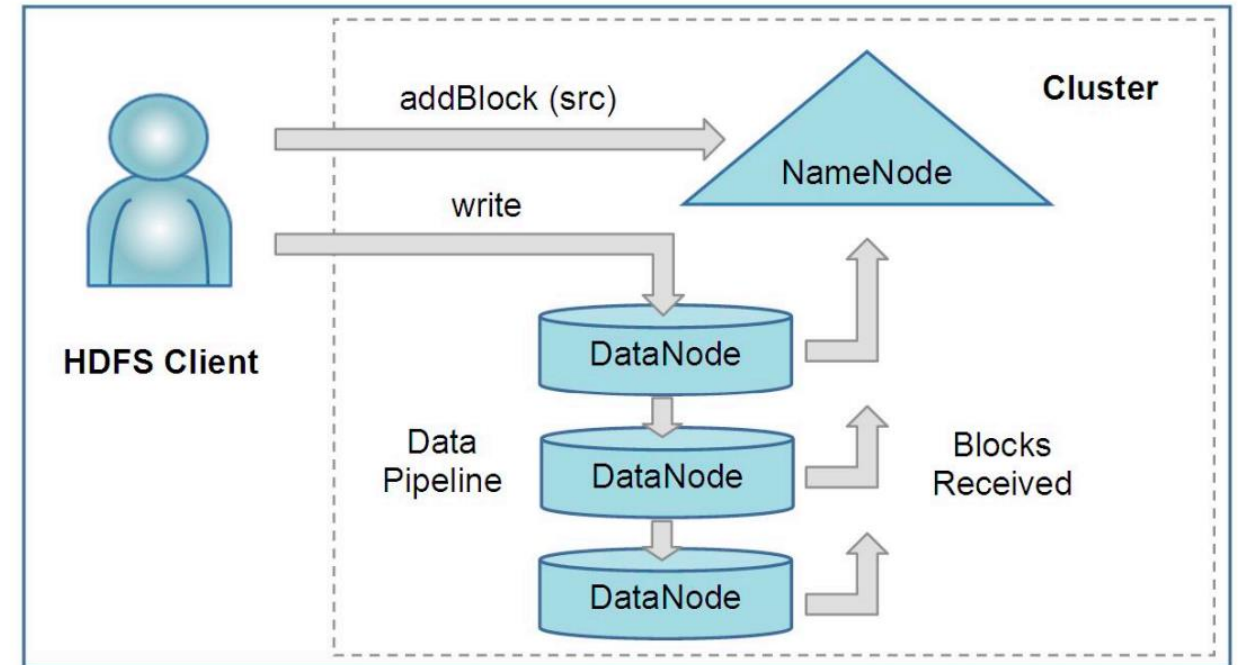
- 5000 GB storage, 10000 IOPS
- 50 Devices
- What do we do with the extra 40000 IOPS?

# Data Warehouse

- Storage for data analytics (map-reduce tables, snapshots of social graph, etc.)
- Read/Write throughput prioritized over latency (Batch processing)
- Files in a same directory are likely to be accessed parallelly

# Hadoop Distributed File System(HDFS)

- NameNode to store and serve metadata
- DataNode to serve read/writes
- **Issue?**
  - HDFS clusters are limited in size because of single NameNode
  - Data needs to be distributed among multiple clusters
  - 2D Bin Packing (NP-Hard) -
    - Cluster capacity
    - Cluster throughput
- **How to scale?**



Source: <https://pages.cs.wisc.edu/~akella/CS838/F16/838-CloudPapers/hdfs.pdf>

# Challenges

## 1. Scale to exabyte scale

- How to store the large amount of metadata?
- How to serve the metadata effectively without being a bottleneck?

## 2. Performance isolation between tenants

- In a storage system used by a large number of tenants, how to prevent one tenant's performance from being affected by others while enabling surplus resource sharing?

## 3. Tenant-specific optimizations

- In a generalized filesystem, how to provide tenants with support that meets the guarantees typically offered by specialized filesystems?

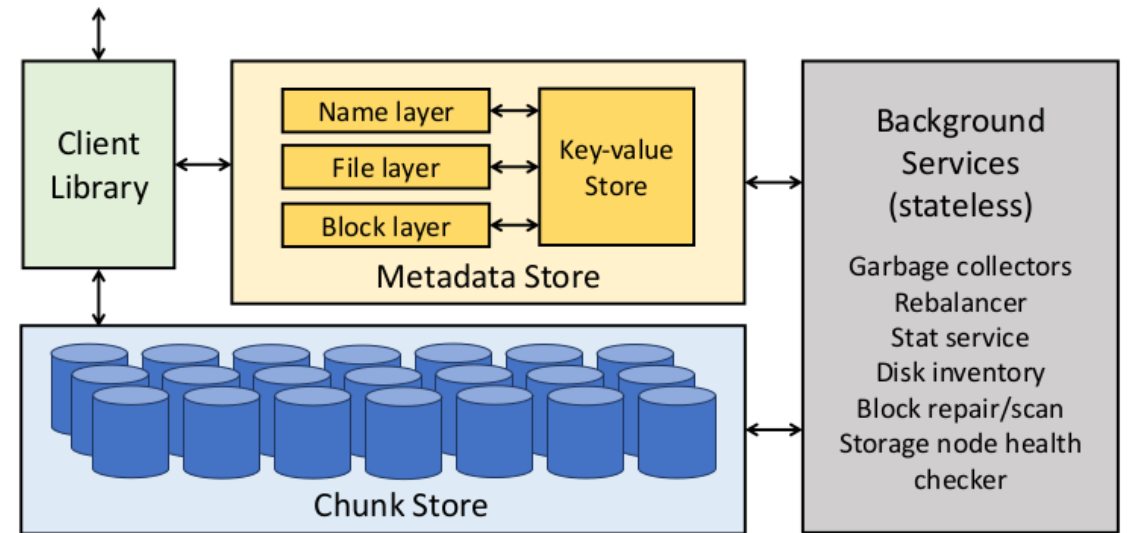


# Some existing Systems

- Federated HDFS and Windows Azure Storage (WAS)
  - **Merge multiple smaller storage clusters into larger clusters**
    - Multiple independent namespaces but Data nodes are shared – Still bin-packing at namespace level (**In which namespace to put what data?**)
- Ceph and Fault-tolerant Distributed Storage(FDS)
  - **Hash Data object to get data location** – Increase the Range of hash function to scale
    - Hash function needs to be updated on each data relocation(Too costly in large systems)

# Tectonic Cluster

- Storage Nodes
- Metadata Nodes
- Stateless Nodes for background operations
- Client library interacts with tectonic



**Figure 2: Tectonic architecture.** Arrows indicate network calls. Tectonic stores filesystem metadata in a key-value store. Apart from the Chunk and Metadata Stores, all components are stateless.

Source: <https://www.usenix.org/system/files/fast21-pan.pdf>

# Chunk Store

- Files divided into blocks and blocks are divided into chunks
- Abstraction created by the client library using metadata
- Chunks are stored as files in cluster's storage nodes using XFS
  1. Scalable – Linearly grows with the number of storage nodes
  2. Read/write to storage nodes can be **specialized to tenant's performance needs**
- Tenants can choose how they want to store the chunks
  1. Replication
  2. Reed-Solomon Encoding [ RS(n,k): n data blocks, k parity blocks ]

# Metadata Store

| Layer | Key                          | Value                  | Sharded by | Mapping                              |
|-------|------------------------------|------------------------|------------|--------------------------------------|
| Name  | (dir_id, <i>subdirname</i> ) | subdir_info, subdir_id | dir_id     | dir → list of subdirs (expanded)     |
|       | (dir_id, <i>filename</i> )   | file_info, file_id     | dir_id     | dir → list of files (expanded)       |
| File  | (file_id, blk_id)            | blk_info               | file_id    | file → list of blocks (expanded)     |
| Block | blk_id                       | list<disk_id>          | blk_id     | block → list of disks (i.e., chunks) |
|       | (disk_id, blk_id)            | chunk_info             | blk_id     | disk → list of blocks (expanded)     |

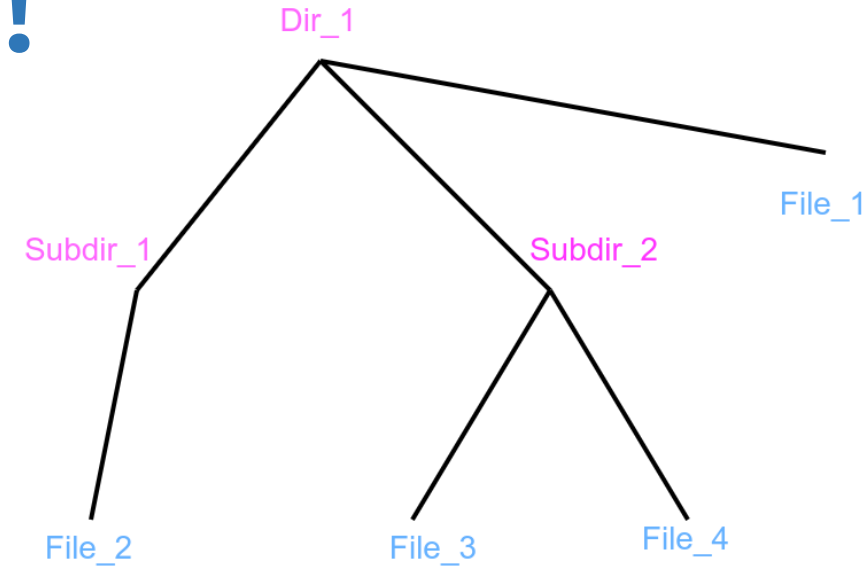
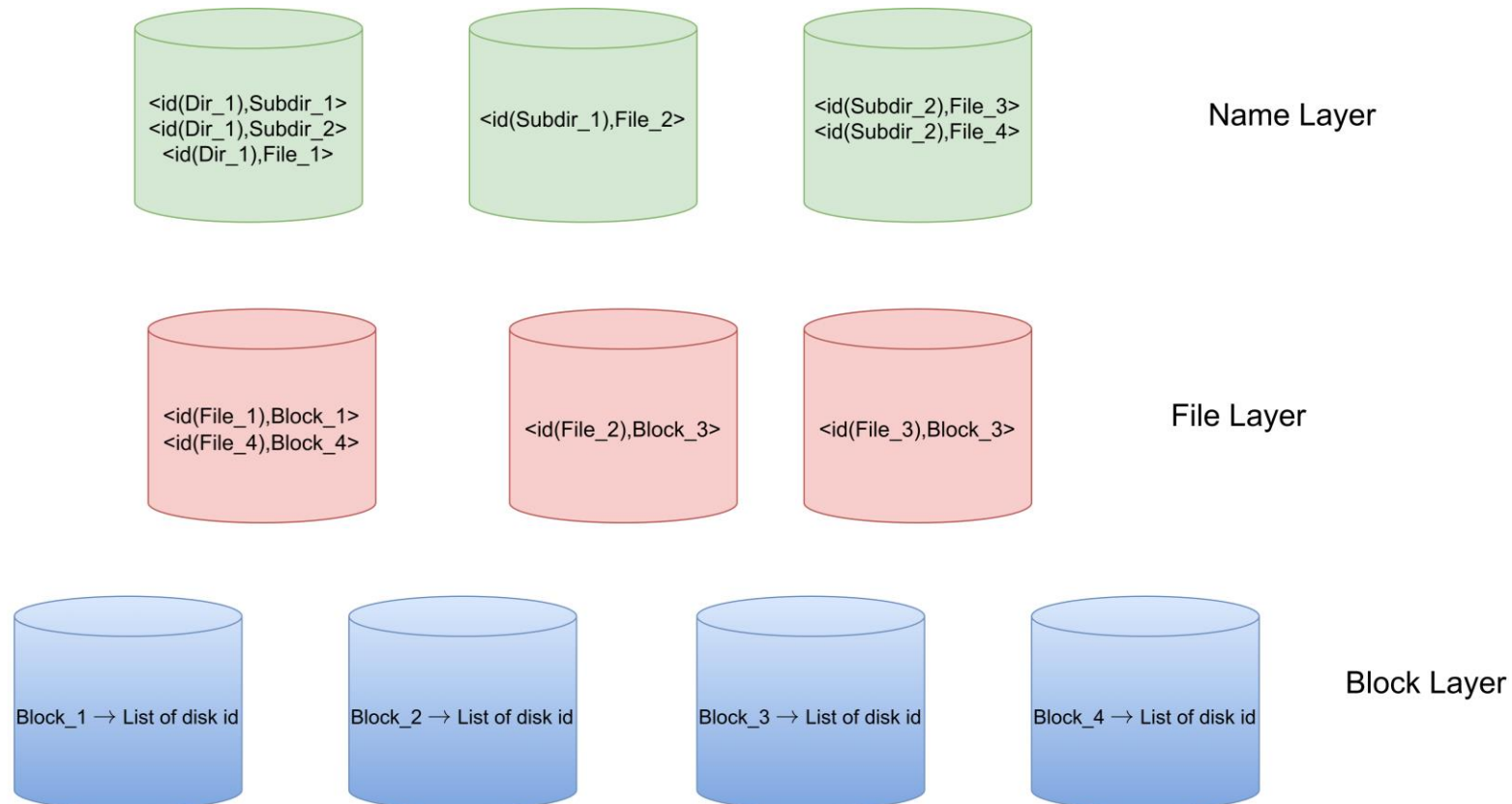
**Table 1: Tectonic’s layered metadata schema.** *dirname* and *filename* are application-exposed strings. *dir\_id*, *file\_id*, and *block\_id* are internal object references. Most mappings are expanded for efficient updating.

- Three metadata layers
  - Name Layer: Directory → Sub-directories/files
  - File Layer: File → Blocks
  - Block Layer: Block → List of disks(chunks), Disk → Block
- Sharded by Key using **Hash Partitioning**
- Stored in **expanded format** (Fast updates)

# Break Metadata into Layers !

1. Expanded Key-Value Pairs
2. Hash Based Partitioning

## Metadata Storage



## Namespace

- Managed by ZippyDB at a shard granularity
- Metadata store nodes internally runs RocksDB and shards are replicated using Paxos
- Advantages
  1. Scalable – Metadata storage grows with number of metadata nodes
  2. Layered metadata approach and hash-partitioning avoids hotspots
- Issue
  1. Higher latency
    - Cache Name and File layer once sealed
    - Block layer cache needs to be updated
  2. No support for atomic cross-directory move operations
  3. No whole directory content listing support. Client Library needs to build it

**Note:** Initially, Name Layer and Block Layer were merged but it turned out to be a hotspot

# Client Library

- Provides filesystem abstraction to applications
- Client library operates read/write operations at chunk granularity

## Single Writer Semantics

- Token provided to client when it opens the file for appending
- Updates to metadata must contain the token
- Last process having opened the file is the only writer
- Tenants needing multiple-writer semantics can build their own serialization semantics on top of tectonic

# Background Services

- Operate on one shard at a time
- Garbage collector
  - Removes inconsistencies between metadata layers
- Rebalance and Repair
  - Identify lost chunks and repair them
  - Uses the reverse index mapping from disks to blocks to scale horizontally



# Multitenancy

- Non-ephemeral resource: Predictable and slow changes and No dynamic adaptation  
Storage Capacity: Pre-allocated capacity
- Ephemeral resource: Real-time automated management
  - IOPS and Metadata Query
- At what granularity to manage?
  - Tenants?
    - Too much generalization – Tenants have different applications with different workloads
  - Applications?
    - Too complex and resource intensive

- **TrafficGroups and TrafficClass**
  - Applications within same Trafficgroup have similar latency and resource requirements
  - TrafficClass – **Gold**(Latency-Sensitive), **Silver**(Normal), **Bronze**(Background applications)
- **Global Resource Sharing**
  - **Aim:**
    - Limit resource usage
    - Share surplus resources effectively
  - Modified Leaky bucket algorithm – High-Performance, Near-Realtime Distributed Counters
    - Increment the counter – If spare capacity, use it
      - Check in other TrafficClass in same tenant
      - Check in other tenants
    - Using other TrafficClass resource – Resulting Traffic Class gets the minimum class

- **Local Resource Sharing at Metadata and Storage Nodes**
  - **Aim:**
    - Avoid it being hotspot
    - Meet latency requirement for Gold class
  - Weighted Round-Robin(WRR): Skip already if quota is used or will exceed if granted
  - Some optimizations:
    - Lower request class can give its turn to higher request if it will have enough time after the higher request gets serviced
    - Limit number of non-Gold traffic in the queue for a disk if there are pending Gold requests
    - **Disks may re-arrange requests!** Stop scheduling non-gold requests if some gold request is pending for some threshold amount of time
- **Note:** Resource sharing need not be done again at Nodes because you have already done it globally

# Access Control

- Token Based Access Control Mechanism
- Each Layer give the token for the next layer
- Tokens are piggybacked

# Tenant Specific Optimizations – Client Library Driven Design

- **Data Warehouse Write Optimizations** – Read only after complete write semantics
  1. **Asynchronous writes:**
    - Applications themselves buffer writes until block size data available
    - RS-encode (less memory and network overhead)
    - Write the chunks in parallel
    - No inconsistency since metadata only update after complete write
  2. **Hedged Quorum Writes:**
    - First send reservation requests to more than the numbers of nodes needed
    - Don't wait for successful completion of writes to all the nodes. Wait only until majority of the blocks have been written and then reply to client.
    - Improves 99th percentile latency

- **Blob Storage Optimizations –**
  - 1. Consistent partial block appends:**
    - Wait only till quorum size appends
    - Only block creator can create append
    - Metadata updated with post-append block size and checksum
  - 2. Reencoding blocks for storage efficiency:**
    - Enabled by Client Library-driven design. Generally filesystems need you to preconfigure the way you want to store files.
    - From replicated to Reed-Solomon(RS) encoding

# Copysets -

Cidon et al. Copysets: Reducing the Frequency of Data Loss in Cloud Storage, Stanford University.

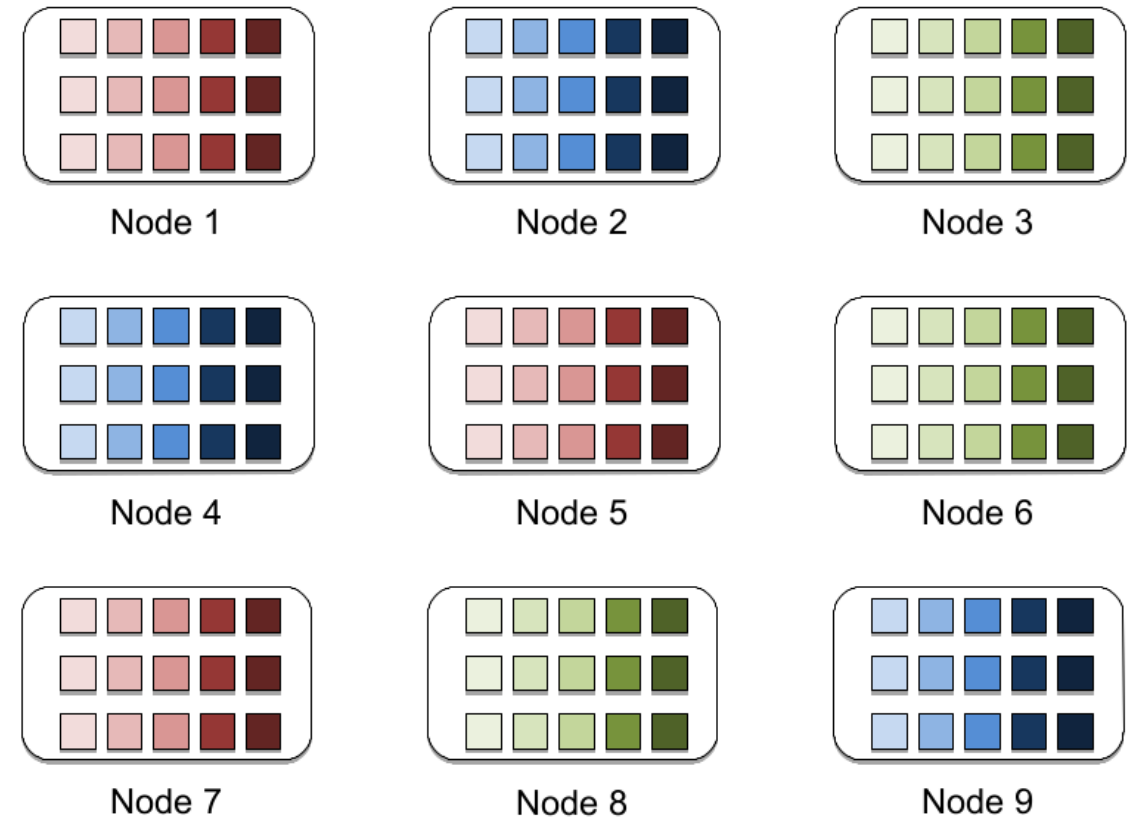
- Set of nodes for which simultaneous crash will lead to data loss
- What if we put copies of data in randomly chosen storage nodes?



- Huge number of copysets
- Data is lost if any three of the nodes go down

Source: <https://www.usenix.org/conference/atc13/technical-sessions/presentation/cidon>

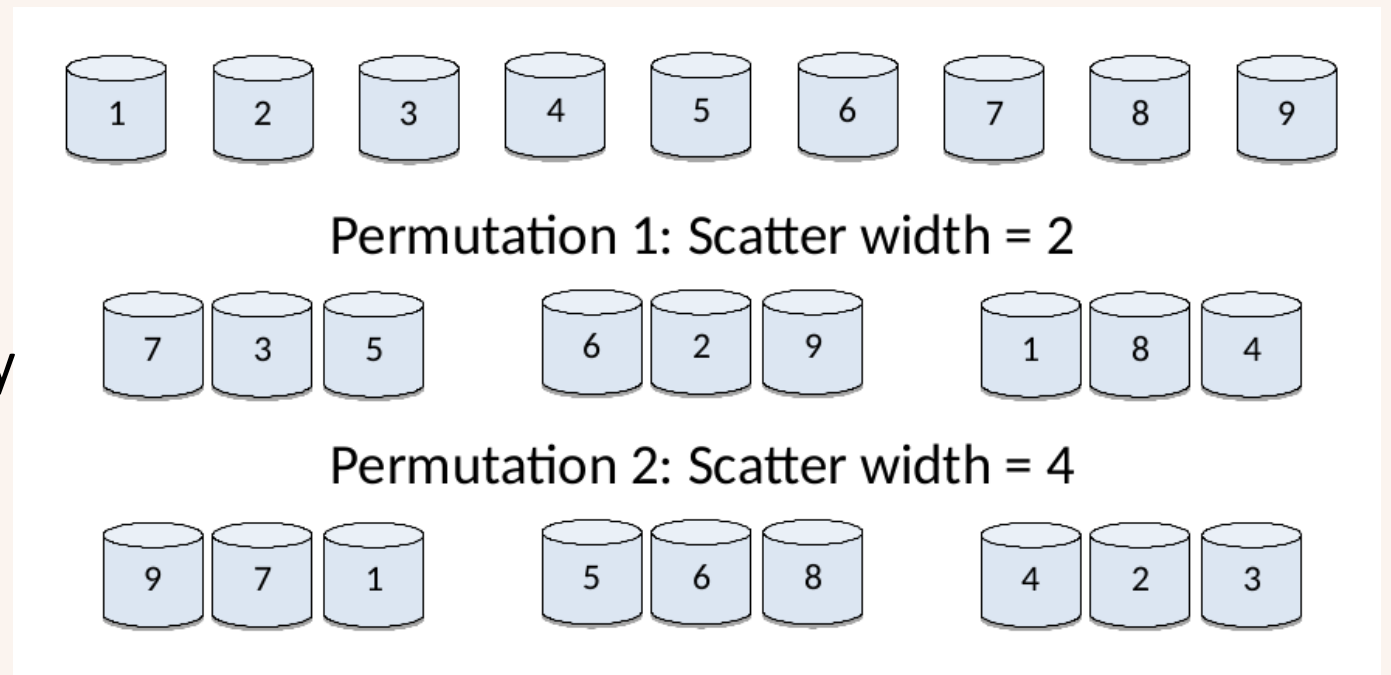
- Only 3 copysets - {1,5,7}, {2,4,9}, {3,6,8}
- Probability of data loss is very low
- **What is the issue?**
- We lose data very less frequently, but we would lose large amount of data
- Reconstruction can be done only from the fellow nodes in the copyset



Source: <https://www.usenix.org/conference/atc13/technical-sessions/presentation/cidon>



- Scatter Width – Number of nodes from which reconstruction can be done
- Create random permutations and group consecutive disks in a copyset
- While writing a block, choose a primary node randomly
- Choose a permutation corresponding to a block id, and use the copyset for the primary node



Source: <https://www.usenix.org/conference/atc13/technical-sessions/presentation/cidon>

- Reconstruction load is now distributed
- In tectonic, the block layer and the rebalancer services together attempt to maintain a fixed copyset count.

# Tectonic in Production: Observations and Lessons

- Performance comparable to specialized filesystems
- Data warehouse request spikes handled by surplus resources
- Only name layer metadata was seen to be hotspot (1% time)
  - Solution: Master uses the metadata using list API and informs the worker nodes
- Client Library access data directly:
  - Alternative design would use a frontend proxy – Avoid an extra network node hop
  - But bugs in library become bug in application
- Reconstruction Storm:
  - Prevent reconstruction to 10%(tuned) of total reads

# Tradeoff

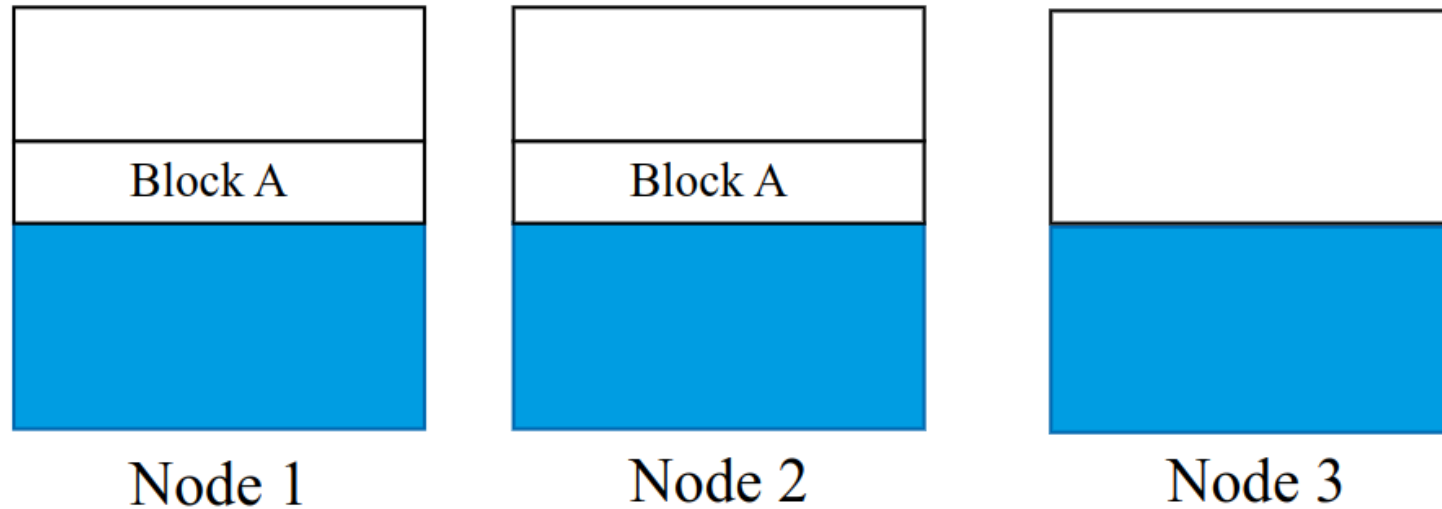
- Higher Metadata Latency
  - As such no direct solution
  - Applications need to adjust how certain metadata operations are handled
    - Parallelize if possible
- Hash partitioned metadata – avoids hotspot
  - No recursive list API – Client needs to build a wrapper for it
    - You don't have updated directory usage information (updated periodically)
      - Commands like du(directory utilization) not available

# Does every Application in Meta use Tectonic?

- Not every service at Meta use Tectonic (Graph storage, Key-Value Storage)
- Many applications communicate with tenants which use Tectonic underneath.
  - Not useful to design a client library for each application

# Observations

- Token based single writer semantics:
  - May lead to deadlock (Alternating retry)
- The eventual consistency model for the partial block appends has performance/consistency issues



A series of thin, light brown lines forming an abstract, overlapping geometric pattern on the left side of the slide. The lines intersect to create various polygonal shapes, some of which are filled with a very light brown color.

# THANK YOU